

Lab - Document Enterprise Cybersecurity Issues

Objectives

Part 1: Record your assessment of Athena's cybersecurity issues.

Part 2: Record the different types of assets owned by Athena.

Part 3: List the threats for each asset type.

Part 4: Recommend mitigation techniques to address each threat.

Note: This lab assumes you have basic knowledge of vulnerabilities, threats, and mitigation techniques.

Scenario

Athena Learning Incorporated is an educational service provider. Athena has two major lines of business: course content creation and online learning services. Athena creates learning content and hosts learning content. Athena also provides internet sales services that enable its partners to charge their students to attend their courses.

Athena employs about 100 people in its headquarters office, and about 5 people each in its London and Singapore offices. Because it provides content and delivery services globally, Athena must comply with diverse privacy and security standards.

Athena serves as custodian for its own content and content that belongs to its partners. That content includes text, graphic, video, and interactive assets. This content is the essential intellectual property of the company. It also manages student account information including student registration, authentication, records, and payment information. Athena manages its own SQL databases, some of which are connected to web portals.

The Athena network consists of mostly MS Windows and Apple IOS clients with a mix of Microsoft and Linux servers to store business and employee records, learning content assets, and financial information, including customer data. The hosts include various PC brands and models of varying age. Different versions of operating systems are in use. Athena uses cloud services to deliver courses to the public, but must house assembled courses on the internal network for creation and editing. When the courses become available, they are mirrored to the cloud. Employees are permitted to use their personal phones and tablets for work. In addition, some employees work from home, but require full network access to do so. Athena also hosts its own DNS, email, and intranet services.

Athena employees use common office application software, custom applications, and tools that have been created internally.

Athena provides access to parts of its internal network to its partners through a secure web portal. Clients are able to preview their course content and deliver course assets to Athena for assembly in the Athena learning management system. Students interact with the cloud-managed learning platform through their web account logins.

In this lab, you will apply your knowledge of cybersecurity threats and mitigation techniques to a corporate setting. You will read about a business, classify its assets, and then list the potential vulnerabilities and threats that the business faces. Finally, you will recommend threat mitigation measures for the threats that you identify.

Required Resources

= Devices with internet access

Instructions

Part 1: Record your assessment of Athena's cybersecurity issues.

Study the **Scenario** above about Athena Learning Incorporated. Focus on identifying the data, software, hardware, and network assets that need to be protected to ensure that company is not impacted by various types of threats that have been discussed in the course so far.

Use the tables below to record your answers.

Information/Data Assets	Threats	Mitigation
Student personally identifiable information on servers	Data theft of protected information, identity theft	Encrypt stored data
Student sales information	Identity theft, theft of credit card information	Encrypt store data
Learning content on course delivery website	Unauthorized access, data theft, alteration of content	Strong authentication, hardening of web application
Customer assets during upload to Apollo	Data tampering, data theft	VPN, hashtag of assets
Data and services required by work-at-home employees	Data theft, data tampering	VPN

Software Assets	Threats	Mitigation
Host Operating Systems	Malware, phishing, malicious websites, security vulnerabilities	Antivirus software, block access to known malicious sites, user security training, patching.
Office Applications	Exploits of unpatched vulnerabilities, various application attacks	Patching, security policies regarding use of unauthorized software.
SQL Databases	Data entry errors, XML or SQL injection	Data input validation
Web Server Software	Vulnerable Server Software, Cross-site scripting	Patching
In-House Applications	Various Application Exploits	Secure software development , strong input validation.

Physical Assets	Threats	Mitigation
Desktop PCs's	Power interruption, hard drive failure, other physical damage, control access to facilities.	Power protection, badge based or biometric access control
Laptop PCS	Loss, theft, or damage, hard drive failure, other physical damage	Antivirus, host-based firewall, regular data backups, drive encryption, physical access control
File Servers	Power interruption, hard drive failure, physical damage	Power protection, automated backups, physically secure server room, redundant servers.
Networking Equipment	Power interruption, physical damage, unauthorized administrative access	Power protection , Physically secure wiring closets and equipment locations

Network Assets	Threats	Mitigation
IP Services (DNS,IMAP,DHCP)	DNS spoofing address spoofing	Next- generation firewalls capable of deep packet inspection.
Connection to ISP	DoS, DDoS	Block external ICMP packets with firewalls
Wired LAN	Man-in-the-middle, unauthorized access, detect unauthorized network scanning.	Strong passwords
Wireless LAN	Man in the middle , unauthorized access, rogue access points	Use strong authentication and encryption , access point placement, use tools to detect rogue access points , VPN for remote wireless users.

Part 2: Record the different types of assets owned by Athena.

From the information in the **Scenario**, and your knowledge of business in general, fill in the first columns of each table with the relevant assets that are owned by Athena. You should have at least three entries in each table.

The different types of assets are defined as follows:

- = **Information/Data Assets** - any data that is used by the company, in any of the three states of data. This data could be Athena's business data, Athena's learning content, student sales and learning data, or partner data.

- = **Software Assets** - any software that is used by Athena, including commercial business applications, operating systems, server software, database software, and custom software.
- = **Physical Assets** - the physical devices, equipment, and other property that are used by Athena in the course of their business.
- = **Network Assets** - the types of networks and network connections that are hosted or used by Athena in the course of its business.

Part 3: List the threats for each asset type.

- a. Review the information that you have learned in this pathway regarding vulnerabilities and threats.

What is the difference between a threat and a vulnerability? Vulnerabilities are weaknesses or characteristics of an asset that can result in damage to or loss of those assets. Threats are the possible actions or events that exploit vulnerabilities. Threats can be posed by people or nature.

- b. Complete the second column of the table with threats that could exploit vulnerabilities for each asset that you listed. There is usually more than one threat to each asset.

Part 4: Recommend mitigation techniques to address each threat.

Review the information that you have learned so far about ways to mitigate various cybersecurity threats. Complete the third column of the table with mitigation techniques that can be done to avoid or limit the damage caused by each potential threat.

Reflection

1. Why is it useful to categorize assets when identifying threats and mitigation techniques? Classifying assets by type helps to organize thinking around what threats may exist. Otherwise, there are so many assets that is difficult to get started with the analysis.
2. Do some threats have the same or similar mitigation measures? Why is it important to note this? Yes, some threats can be mitigated using the same means. A system of updating host and server software can help to mitigate threats to customer and company data.
3. What have you learned about the application of knowledge of cybersecurity threats and mitigation techniques to the context of a simulated organization? I learned that there is no single solution to protecting assets from the wide range of threats that exist in the world today.